

Ihsan Salari

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Education

BASc Honors Electrical Engineering, Co-op *University of Waterloo* GPA: 89% **Waterloo, ON** 2024 - 2029

Qualifications

- **RF/Power:** class-DE GaN inverters, tens-of-MHz RF plasma generation, V-I probes, 3.5MHz GaN dc-dc
- **High-speed/SI:** 224G PAM4 SerDes, PCIe Gen 5, 102.4T switches, CPO/LPO optics
- **CAD:** Cadence Allegro/OrCAD, Altium, KiCad, LTspice, HFSS, HyperLynx, ADS
- **Lab:** 70GHz VNA, TDR, Bode Plots, Logic Analyzer, SMD rework
- **Programming:** C/C++, Python, MATLAB, SPICE, CUDA; STM32, AMD/Xilinx Zynq SoC

Experience

Hardware Engineering Intern *Fab2 (formerly Atomic Semi)* **San Francisco, CA** 05/2026 - present

- Designed **200W class-DE GaN sine inverter** at **tens of MHz** for **RF plasma generation** in a single day; carried it from schematic to first plasma in three weeks.
- Designed **V-I probe** for **tens-of-MHz RF** power delivery, resolving voltage, current and phase at the load.
- Built RF amplifier and plasma-generation power system hardware for **fast-turn semiconductor fabrication tools**.
- Developed **AI-accelerated PCB design workflow**; built and applied **AI tools** to ramp on unfamiliar RF, plasma and power-electronics concepts the designs required.

Signal Integrity and Hardware Design *Arista Networks, Inc.* **Santa Clara, CA** 09/2025 - 12/2025

- Ensured SI on highest priority **102.4T** switch by simulating **224G PAM4 SerDes** and **PCIe Gen 5** differential pairs using **Hyperlynx/HFSS/ADS** and advised layout to meet loss budgets, reduce crosstalk, reflections, and resonances, and meet DFM.
- Unblocked fabout of two most-advanced 102.4T switches through mission-critical **224G SI debug** using **70GHz VNAs**; ran and analyzed **TDRs** to pinpoint source of discontinuities, quantify skew, and determine impedance variation due to microstrip section.
- Owned end-to-end redesign of **PCIe Gen 5 SSD bridge** card from schematic changes to layout instructions to SI to fabout; optimized via impedance (antipads, NFPs, spacing, diameters) to meet -20 dB return loss at Nyquist.
- Fast-tracked selection of compact, highly efficient PMICs for **1.6T LRO OSFP** and **CPO optics**, shaving weeks off schedule.
- Caught fatal ethernet magnetics saturation issue on Eth/RS-232 multiplexing project by implementing lab and SPICE sim setups.

Power Electronics Designer *aiRadar, Inc.* **Vancouver, BC** 01/2025 - 05/2025

- Spearheaded end-to-end redesign of cutting-edge **3.5MHz GaN** wide input/output multi-stage dc-to-dc converter for advanced multi-beam sonar, including research, topology selection, simulation, firmware development, bringup, and testing.
- Proposed and validated dc-to-dc converter topologies using **LTspice** AC/DC/transient analysis accounting for parasitics; prototyped on GaN FET eval kits and STM32 dev boards for initial testing and PID tuning.
- Implemented **STM32** firmware with voltage-fed **PID control**, live telemetry, and extensive **UART** command interface; developed and executed board bring-up and test plan.

Technical Projects

USB-C Trigger Board 08/2025

- End-to-end design of **140W USB-C Power Delivery (PD)** trigger board with Extended Power Range (EPR) serial interface.
- Implemented solution based on USB-C PD IC with I2C configuration via onboard STM32 hosting custom negotiation firmware.
- Designed **6-layer PCB** optimizing return paths, power trace dimensions and thermals, and subsequently reflowed prototypes.

Electrohydrodynamic (Ionic) Propulsion 09/2022 - 10/2023

- Designed and constructed **50kV 100W** flyback converter with Cockcroft-Walton voltage multiplier simulated in **LTspice**.
- Designed and built working electrohydrodynamic (ionic) thruster that achieved wind speeds of 1.5m/s and thrust of 40mN.
- Building on **recent MIT research**, authored a paper deriving optimal electrode spacing for parallel multi-electrode ionic thrusters, validated experimentally.